

Computational Model of Planning Depth and Reaction Time

1 Computational Model

We developed a computational model to simulate reaction times (RTs) in a hierarchical three-step sequential decision task. The model captures the cognitive dynamics of planning initiation, execution, and learning as a function of planning depth.

1.1 Task Structure

Participants made three sequential binary decisions ($d \in \{1, 2, 3\}$) to reach a goal state in a decision tree. Planning depth ($\delta \in \{1, 2, 3\}$) determined when participants needed to initiate planning:

- **Depth 3** ($\delta = 3$): Plan all three actions at decision 1. Only one possible route through the tree ($|\mathcal{P}_3| = 1$).
- **Depth 2** ($\delta = 2$): Plan two actions starting at decision 2. Two possible routes ($|\mathcal{P}_2| = 2$).
- **Depth 1** ($\delta = 1$): Plan one action at decision 3. Three possible routes ($|\mathcal{P}_1| = 3$).

The planning initiation decision d_{plan} was determined by depth:

$$d_{\text{plan}}(\delta) = 4 - \delta \tag{1}$$

Goals could change between blocks of trials. When a goal changed, a binary indicator s_t was set to 1 for that trial, otherwise 0.

1.2 Path Assignment and Experience Tracking

For each trial t , we assigned a specific path p_t through the decision tree based on the planning depth. Path assignment was stochastic:

$$p_t \sim \text{Uniform}\{0, 1, \dots, |\mathcal{P}_{\delta_t}| - 1\} \tag{2}$$

We tracked path-specific experience $E_p(t)$ for path p across the entire experiment. Critically, path experience **persisted across goal switches**—participants did not forget path-specific learning when goals changed. Experience evolved according to:

$$E_p(t) = \begin{cases} \lambda_{\delta_t} \cdot E_p(t-1) + 1 & \text{if } p \neq p_{t-1} \\ \lambda_{\delta_t} \cdot E_p(t-1) + 1 + \beta & \text{if } p = p_{t-1} \end{cases} \quad (3)$$

where λ_δ is the depth-dependent forgetting rate and β is a successive repetition bonus. For all paths $p' \neq p_t$ (unused on trial t):

$$E_{p'}(t) = \lambda_{\delta_t} \cdot E_{p'}(t-1) \quad (4)$$

Note that forgetting is applied every trial to all paths, but the forgetting rate depends on the current trial's depth. Forgetting rates were parameterized to reflect that paths practiced more frequently (fewer total paths at a given depth) should have stronger memory:

$$\lambda_\delta = \begin{cases} 0.98 & \text{if } \delta = 3 \\ 0.95 & \text{if } \delta = 2 \\ 0.90 & \text{if } \delta = 1 \end{cases} \quad (5)$$

1.3 Reaction Time Components

The predicted reaction time $\text{RT}_{t,d}$ for trial t at decision d was modeled as a sum of cognitive components:

$$\text{RT}_{t,d} = C_{\text{base}} + C_{\text{goal}}(t, d) + C_{\text{plan}}(t, d) - C_{\text{exec}}(t, d) - C_{\text{comp}}(t, d) + C_{\text{switch}}(t) + \epsilon_{t,d} \quad (6)$$

where each component is defined below, and $\epsilon_{t,d} \sim \mathcal{N}(0, \sigma^2)$ represents Gaussian noise.

1.3.1 Base Time

A constant baseline for all decisions:

$$C_{\text{base}} = \tau_0 \quad (7)$$

We set $\tau_0 = 0$ as a reference point, with all other components scaled relative to this baseline.

1.3.2 Goal Processing Cost with Learning

At decision 1, participants identified which goal to pursue. This processing cost decreased exponentially with overall experience across the entire experiment:

$$C_{\text{goal}}(t, d) = \begin{cases} \alpha_{\text{goal}} \cdot \exp(-\gamma_{\text{goal}} \cdot (t-1)) & \text{if } d = 1 \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

where α_{goal} is the initial goal processing cost, γ_{goal} is the learning rate, and t is the overall trial number. Critically, this learning **persists across goal switches**—participants become progressively faster at goal processing throughout the experiment, regardless of which specific goal they are pursuing.

1.3.3 Planning Initiation Cost

When planning was initiated (at decision d_{plan}), participants incurred a cost that scaled with the number of actions to plan:

$$C_{\text{plan}}(t, d) = \begin{cases} \alpha_{\text{init}} + \alpha_{\text{step}} \cdot (4 - d_{\text{plan}}) & \text{if } d = d_{\text{plan}}(\delta_t) \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

where α_{init} is the base cost of initiating planning, α_{step} is the cost per action to plan, and $(4 - d_{\text{plan}})$ equals the number of steps to plan (ranging from 1 to 3).

1.3.4 Execution Discount

After planning was initiated, subsequent decisions benefited from a pre-computed plan, resulting in faster execution. This discount increased with path experience:

$$C_{\text{exec}}(t, d) = \begin{cases} \alpha_{\text{exec}} \cdot (d - d_{\text{plan}}) \cdot (1 + \beta_{\text{exec}} \cdot \ln(1 + E_{p_t}(t))) & \text{if } d > d_{\text{plan}} \\ 0 & \text{otherwise} \end{cases} \quad (10)$$

where α_{exec} is the base execution discount per step, β_{exec} scales the experience boost, and $E_{p_t}(t)$ is the current path experience.

1.3.5 Path Compression

Repeated execution of the same path led to compression—a speeding of RTs due to proceduralization. Compression was depth-dependent, with stronger effects for paths practiced more frequently:

$$C_{\text{comp}}(t, d) = \begin{cases} \alpha_{\text{comp}} \cdot \kappa_{\delta_t} \cdot \ln(1 + E_{p_t}(t)) & \text{if } d \geq d_{\text{plan}} \\ 0 & \text{otherwise} \end{cases} \quad (11)$$

where α_{comp} is the base compression coefficient, and κ_{δ} is a depth-dependent scaling factor:

$$\kappa_{\delta} = \begin{cases} 1.0 & \text{if } \delta = 3 \\ 0.5 & \text{if } \delta = 2 \\ 1/3 & \text{if } \delta = 1 \end{cases} \quad (12)$$

The logarithmic form $\ln(1 + E)$ captures diminishing returns of practice, consistent with power-law learning [1].

1.3.6 Goal Switch Cost

When the goal changed from one trial to the next, participants incurred a transient reorientation cost:

$$C_{\text{switch}}(t) = \begin{cases} \alpha_{\text{switch}} & \text{if } s_t = 1 \\ 0 & \text{otherwise} \end{cases} \quad (13)$$

Importantly, goal switches only added this transient cost—they did not reset path experience (Equations 3–4) or goal processing learning (Equation 8). This reflects the assumption that low-level procedural and perceptual learning persists even when task goals change.

1.3.7 Noise and Minimum RT

We added Gaussian noise to simulate trial-by-trial variability:

$$\epsilon_{t,d} \sim \mathcal{N}(0, \sigma^2) \quad (14)$$

and enforced a minimum physiologically plausible RT:

$$\text{RT}_{t,d}^{\text{final}} = \max(\text{RT}_{t,d}, \text{RT}_{\min}) \quad (15)$$

1.4 Model Parameters

Table 1 summarizes all model parameters and their values.

Parameter	Description	Value
τ_0	Base time	0.00 s
α_{goal}	Initial goal processing cost	0.50 s
γ_{goal}	Goal learning rate	0.30
α_{init}	Planning initiation cost	0.50 s
α_{step}	Cost per action to plan	0.15 s
α_{exec}	Base execution discount	0.20 s
β_{exec}	Execution experience boost	0.10
α_{comp}	Base compression coefficient	0.90 s
κ_3	Depth-3 compression scale	1.00
κ_2	Depth-2 compression scale	0.50
κ_1	Depth-1 compression scale	0.33
λ_3	Depth-3 forgetting rate	0.98
λ_2	Depth-2 forgetting rate	0.95
λ_1	Depth-1 forgetting rate	0.90
β	Successive repetition bonus	1.00
α_{switch}	Goal switch cost	0.20 s
σ	RT noise standard deviation	0.08 s
$\text{RT}_{\min}$	Minimum RT	0.05 s

1.5 Model Predictions

The model generates several key predictions:

1. **Planning initiation cost by depth:** At the planning decision, RT should increase with depth: $RT|_{\delta=3,d=1} > RT|_{\delta=2,d=2} > RT|_{\delta=1,d=3}$ due to the $(4 - d_{\text{plan}})$ term in C_{plan} .
2. **Depth-dependent learning rates:** Compression should be strongest for depth 3 (one path, $\kappa_3 = 1.0$) and weakest for depth 1 (three paths, $\kappa_1 = 0.33$).
3. **Goal processing learning:** Decision 1 RTs should decrease exponentially across the entire experiment, with the rate of decrease governed by γ_{goal} . This learning should not reset when goals switch.
4. **Execution speeding:** RTs should decrease for decisions after planning initiation, with the magnitude of decrease scaling with path experience via C_{exec} and C_{comp} .
5. **Transient switch costs:** Transitioning to a new goal should produce transiently elevated RTs (duration of one trial) due to C_{switch} , but all accumulated learning should persist.
6. **Continuous learning:** Path compression and goal processing should show monotonic improvement throughout the experiment, uninterrupted by goal switches.

1.6 Normalization

For visualization and comparison across subjects, we z-scored the simulated RTs within each subject:

$$RT_{t,d}^{\text{norm}} = \frac{RT_{t,d} - \mu_{\text{sub}}}{\sigma_{\text{sub}}} \quad (16)$$

where μ_{sub} and σ_{sub} are the mean and standard deviation of all RTs for that subject.

1.7 Implementation

The model was implemented in Python 3.9 using NumPy 1.24 and Pandas 2.0. For each simulated subject, we:

1. Initialized path experience counters $E_p(0) = 0$ for all paths p .
2. Iterated through trials in chronological order.
3. Updated path experience according to Equations (3)–(4), with forgetting applied continuously regardless of goal switches.
4. Computed RT components for each decision according to Equations (7)–(13).
5. Added noise and enforced minimum RT (Equations (14)–(15)).
6. Normalized RTs within subject (Equation (16)).

A critical implementation detail: path experience and goal processing learning accumulated across the entire experiment without resetting at goal boundaries, reflecting the assumption that procedural and perceptual learning persist independently of higher-level goal representations.

References

- [1] Newell, A., & Rosenbloom, P. S. (1981). Mechanisms of skill acquisition and the law of practice. *Cognitive skills and their acquisition*, 1, 1–55.